Transition of Care AI-Assisted Proof of Concept Leveraging the Clinical Health AI-augmented Records Toolkit

Assembling structured and unstructured data into clinically relevant information using generative artificial intelligence (AI), HL7® FHIR®, and the PACIO Transitions of Care Implementation Guide.

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Introduction & Objectives

Clinical problem

Post-acute transitions are often hindered by incomplete, fragmented, and inconsistently represented handoff information. Clinical data are scattered across electronic health record (EHR) fields, notes, and documents, resulting in duplication, conflicting information, unclear data provenance, and missing context. Receiving teams must often reconstruct the patient's clinical story with inadequate information.

The use case is also shaped by a handoff information-needs gap: the discharging team provides the information it believes the next team needs, while the receiving team may require a different subset, level of detail, or clinical context to continue care. When the needed information is distributed across structured fields and narrative sources, the receiving team may need to manually reconcile the patient story under time pressure.

Why standards alone are insufficient

The PACIO Project and HL7® Fast Healthcare Interoperability Resources (FHIR®) enable standardized data exchange but do not inherently support clinical interpretation. They do not filter for relevance, reconcile conflicting or duplicate data, assess completeness or accuracy, prioritize actionable information, or provide the person-centered view needed for effective care transitions.

Standards establish how data can be represented and exchanged, but applications and workflows still need to perform harmonization, de-duplication, organization, provenance management, and clinician review. This is particularly important when transition information must be assembled from a mixture of structured EHR fields, free-text notes, documents, and other source artifacts.

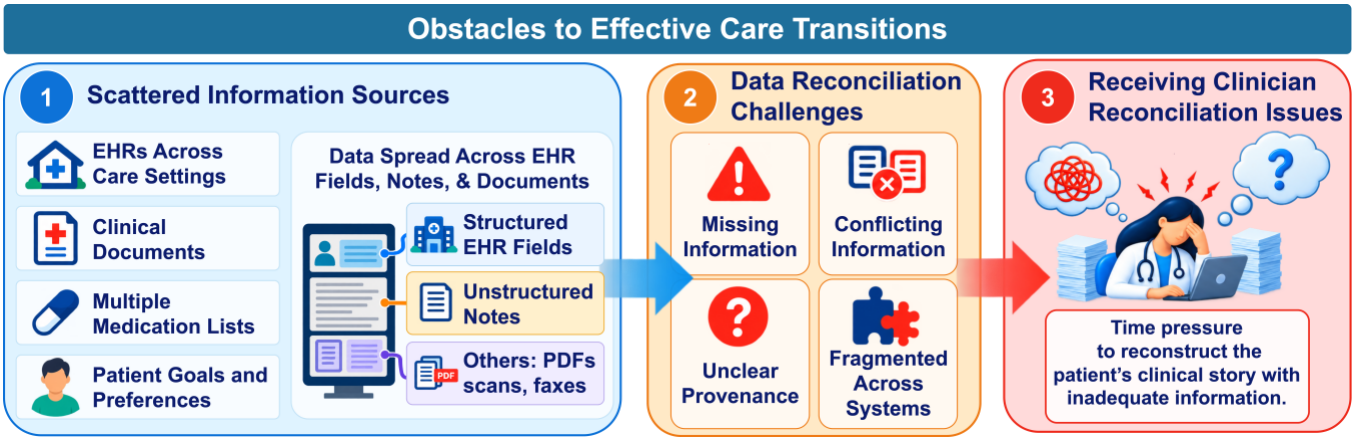


Figure 1. Obstacles to effective care transitions: scattered information sources, reconciliation challenges, and receiving clinician burden.

Objective

To address these gaps, we demonstrate the **Transition of Care AI-Assisted (ToC-AI)** app, a standards-led, AI-augmented exemplar application that leverages services from the **Clinical Health AI-augmented Records Toolkit (CHART)** to consolidate structured and unstructured data and align outputs with the PACIO Transitions of Care (ToC) Implementation Guide (IG). ToC-AI identifies and organizes relevant data elements, while clinicians retain review and reconciliation authority with visibility into source evidence. The goal is to move beyond data exchange alone by preparing complete, meaningful, timely, and transition-ready information for the next care team.

Materials & Methods

Prototype setting and inputs

Synthetic post-acute care patient scenarios combining structured and free-text clinical records were used to develop and evaluate the prototype. The scenarios were designed to represent the variety of formats and information sources encountered in transition workflows without using real patient data. Development incorporated iterative input from the PACIO ToC Subgroup and drew on FHIR, United States Core Data for Interoperability (USCDI) Version 5, and the Patient Information Quality Improvement (PIQI) Framework as technical references.

Reference terminology and data-quality work included SNOMED CT, LOINC, RxNorm, and International Classification of Diseases, Tenth Revision, Clinical Modification (ICD-10-CM). Broader prototype planning also considered Current Procedural Terminology (CPT) and post-acute assessment content represented in the CMS Data Element Library. The PACIO ToC Implementation Guide served as the primary benchmark for the structure and expected content of the transition output; USCDI v5 supported identification of key interoperable data elements rather than functioning as a complete ToC content checklist.

Prototype systems and architecture

The prototype consisted of two complementary systems:

- **CHART (Clinical Health AI-augmented Records Toolkit**), a reusable, modular AI toolkit that provides services for clinical information retrieval and extraction, analysis, synthesis and harmonization, standards transformation, and data-quality assessment.
- **ToC-AI (Transition of Care AI-Assisted) Application**, a proof-of-concept implementation that used CHART capabilities to orchestrate an end-to-end transitions-of-care workflow encompassing information assembly, clinician review and reconciliation, and preparation of PACIO ToC-aligned FHIR output.

CHART is a standards-led, modular, agent-based architecture in which an **Orchestrator Agent** coordinates the sequence of processing activities, routes information among CHART services, and manages handoffs as data move through the workflow. The orchestrator invokes the appropriate processing services while allowing the underlying capabilities to remain modular and reusable across different healthcare use cases.

CHART consolidates structured and unstructured data into standardized and harmonized healthcare information through complementary services:

- **Data Retrieval and Extraction Agent:** retrieves patient information from multiple structured formats and free text and can combine local and remote data sources to support a more comprehensive patient record.
- **Data Analysis Agent:** identifies applicable USCDI data elements, maps clinical concepts to standard vocabularies, and evaluates data against applicable quality frameworks such as PIQI.
- **Data Synthesis and Harmonization Agent:** reconciles information from multiple sources, including duplicate or conflicting entries, and applies large language model (LLM)-assisted matching to support de-duplication, data ranking, and conflict resolution recommendations.
- **Data Transformation Service:** converts harmonized information into a final report or data product represented in one or more healthcare standards, specifically FHIR for the ToC-AI implementation.

A shared **Data Service Layer** supports most of these processing services by providing common data and reference resources throughout the workflow. Its components include:

- **Staging Data Store:** maintains working patient data and intermediate representations as information is retrieved, analyzed, harmonized, and prepared for downstream transformation.
- **Terminology Model Context Protocol (MCP) Server:** provides access to terminology resources and retains retrieved terminology content for reuse, including SNOMED CT, LOINC, RxNorm, and ICD-10-CM codes.
- **Knowledge Base:** provides reusable reference knowledge used by the processing services, including PIQI, USCDI v5, applicable healthcare data standards, and related implementation guidance.

Together, these shared services allow CHART agents to operate against a common set of staged data, terminology resources, and standards-based knowledge rather than requiring each service to independently retrieve or maintain the same information. The Data Service Layer is connected to most of the CHART processing agents, providing shared context and resources as patient information progresses through the workflow.

These capabilities are designed to be reusable across healthcare interoperability use cases rather than being specific to transitions of care. In the ToC-AI exemplar, CHART services are composed into the transition workflow shown in Figure 2, from source-data processing through clinician adjudication and preparation of PACIO ToC FHIR output.

ToC-AI therefore represents an application of CHART rather than a separate AI toolkit. The architecture separates reusable CHART capabilities and shared data services from the ToC-specific workflow, allowing the same underlying services to be applied to other clinical, quality, and interoperability use cases.

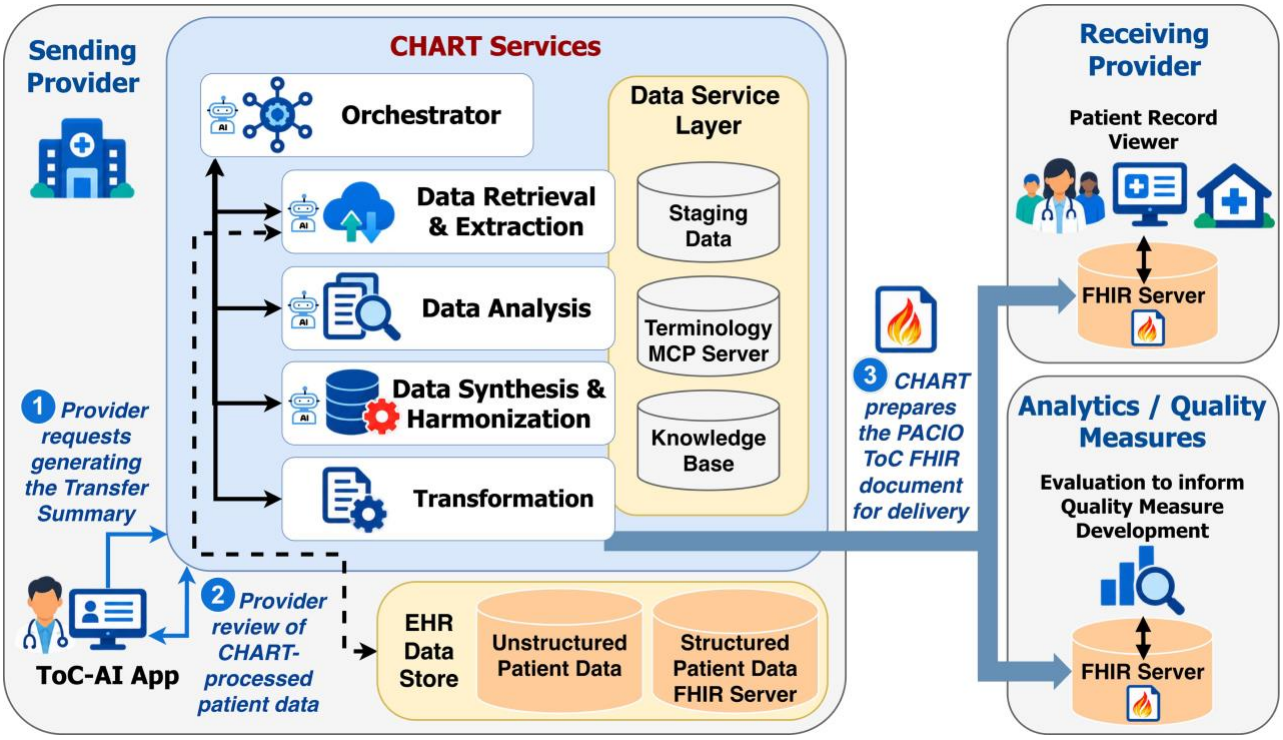


Figure 2. ***Transition of Care AI-Assisted (ToC-AI) App Workflow:*** *A sending provider initiates transfer-summary generation, reviews CHART-processed information and prepares PACIO ToC FHIR output for downstream delivery and evaluation.*

AI-assisted data processing workflow

Within the ToC-AI exemplar, the CHART services described above were applied as an integrated processing workflow using synthetic post-acute care records. The workflow operationalized these services through the sequence of activities below, spanning initial data retrieval, clinician review, and preparation of standards-aligned transition information:

- Queried and ingested structured and unstructured EHR data from synthetic source records.
- Extracted clinically relevant entities and identified candidate USCDI v5 elements in structured fields and free-text content.
- Maintained source context and traceability for data elements as they were processed and prepared for transition-level information generation.
- Normalized terminology using standard vocabularies, including SNOMED CT, LOINC, RxNorm, and ICD-10-CM, as applicable.
- Detected duplicate and conflicting information and generated confidence or data-quality indicators for review.
- Presented flagged information, source evidence, and AI-generated recommendations for clinician review and reconciliation.
- Packaged reconciled content into a PACIO ToC FHIR IG-aligned FHIR document Bundle.

LLM-enabled analysis supported extraction from narrative text, de-duplication, conflict identification, and ranking of candidate information. These outputs were treated as reviewable suggestions rather than authoritative clinical determinations. The design also contemplated flagging expected information gaps against applicable PACIO ToC content expectations so clinicians could add available information or confirm that an expected item was unavailable; completeness remains a future validation measure.

Clinician review and reconciliation

A deliberate human control point for clinician review and reconciliation was built into the workflow before downstream sharing. Rather than treating AI-generated output as final, the ToC-AI workflow presented

candidate transition information, source evidence, identified conflicts, and information gaps for clinician adjudication. Within this review step, clinicians performed the following activities:

- Clinicians reviewed AI-suggested content before downstream sharing.
- Assessed clinical relevance, source support, and appropriateness for inclusion rather than relying on AI to determine clinical accuracy or relevance independently.
- Resolved conflicting or inconsistent information and reviewed duplicate groups with access to source evidence.
- Addressed identified information gaps by adding available missing information or confirming that information was unavailable, when appropriate.
- Determined readiness of the reconciled transition information for downstream sharing.

Source traceability was treated as a key safety and trust feature: AI-assembled content remained linked to originating artifacts so clinicians could inspect the basis for a recommendation. This approach does not solve inadequate source documentation; when necessary clinical information was never documented in the source record, the system cannot reliably create that information and instead should surface the gap for human follow-up.

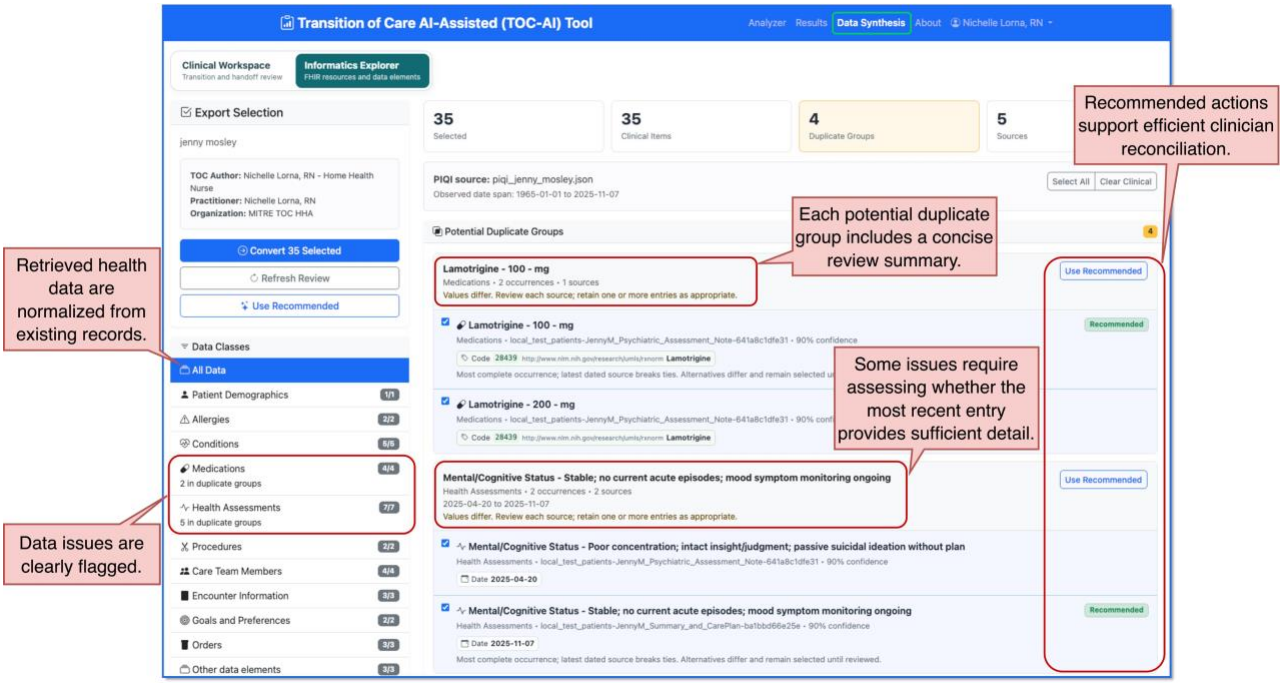


Figure 3. ***Clinician reconciliation workflow:*** *The interface flags data issues, presents source-linked evidence and duplicate summaries, and offers recommendations that remain subject to clinician review before transition information is finalized for sharing.*

Standards and quality approach

Standards and quality controls were applied throughout the ToC-AI workflow. The PACIO ToC Implementation Guide defined the target structure for transition output; USCDI v5 supported identification of key data elements; SNOMED CT, LOINC, RxNorm, and ICD-10-CM supported terminology normalization; and PIQI informed data-quality assessment. These references were operationalized through the CHART analysis, synthesis, and transformation capabilities described above.

In addition to the primary CHART components shown in Figure 2, the implementation included supporting components that operated alongside the main processing workflow and are not depicted in the simplified diagram. The **PIQI-to-FHIR MCP service** provided Model Context Protocol-based support for converting reconciled, PIQI-processed content into FHIR bundles, while the **Quality Evaluation Agent** evaluated extraction quality, PIQI scorecards, LLM judge output, and ToC FHIR conformance. Together, these components supported the transformation and evaluation stages of the workflow, including preparation of

PACIO ToC-aligned FHIR bundles and assessment of the resulting output. In this proof of concept, ToC FHIR conformance assessment refers to alignment with PACIO ToC structures and expectations; it does not establish production-grade interoperability across a broad set of external systems.

Findings and Discussion

Functional proof-of-concept findings

The proof of concept demonstrated technical feasibility of an end-to-end workflow from heterogeneous synthetic source data to clinician-reviewable, standards-aligned transition output. Table 1 summarizes the capabilities demonstrated in the proof of concept and the corresponding measures reserved for future pilots.

Clinical workflow relevance

The prototype consolidated fragmented source data into a unified handoff view, surfaced key transition information and potential risks, and preserved provenance for clinician review. This approach may reduce manual chart-review burden, although that outcome was not measured in the proof of concept. The workflow is designed to help bridge the information-needs gap between sending and receiving care teams by organizing candidate information for human review rather than requiring the receiving team to reconstruct the record from disconnected source artifacts.

Informatics insight

The exploration suggests the value of interoperability extends beyond data movement to semantic structuring, relevance filtering, provenance, and interpretability. By organizing standards-based data into clinically meaningful information, the approach bridges the gap between FHIR exchange and clinical usability. The role of AI in this design is therefore to strengthen the preparation and usability of standards-based exchange, not to replace PACIO IGs, FHIR, clinical terminology standards, or clinician judgment.

Evidence boundary and limitations

Findings are limited to synthetic scenarios and technical feasibility; no real-world deployment, clinical outcomes, or efficiency improvements were established. Table 1 distinguishes what was demonstrated in the proof of concept from what remains to be measured in pilots.

The prototype also depends on the quality and availability of source documentation. It can organize, compare, and flag information that is present and can identify expected gaps, but it cannot reliably supply clinical information that was never documented. Future evaluation should distinguish failures of source documentation from failures of extraction, harmonization, or exchange.

Future exploration

Future evaluation should compare the AI-assisted workflow with baseline processes and distinguish technical performance, clinician experience, interoperability, and downstream transition outcomes.

Real-world deployment will require explicit evaluation of applicable policy, regulatory, privacy, security, data-use, and interoperability-governance requirements. The project has identified governance as part of the path from proof of concept to production exchange, including how data are collected for analysis, how AI-generated or AI-assembled products are reviewed and distributed, and how source provenance and clinician authority are preserved.

Future implementation should also evaluate how the Trusted Exchange Framework and Common Agreement (TEFCA) or other applicable federal interoperability governance frameworks affect the exchange model where relevant. Regulatory and governance alignment should be assessed alongside FHIR conformance, workflow integration, and clinical validation; it was not demonstrated by the synthetic proof of concept.

Table 1 summarizes the evidence boundary for the proof of concept by pairing each demonstrated capability with the corresponding validation or measurement needed in future real-world pilots.

Table 1. Evidence boundary: demonstrated in the proof of concept vs. to be measured in pilots

EVALUATION AREA	DEMONSTRATED IN PROOF OF CONCEPT	TO BE MEASURED IN PILOTS
Data ingestion & source handling	Processed structured and unstructured clinical data from curated synthetic post-acute care scenarios.	Reliability across real EHR, health information exchange, FHIR, file-share, and document-repository inputs, including source variability and connector performance.
Clinical data extraction	Identified clinically relevant information from structured fields and free-text notes.	Extraction accuracy and completeness across representative post-acute care data types and documentation styles.
Standards mapping	Mapped candidate information to key USCDI v5 and PACIO Transitions of Care elements.	Mapping accuracy and completeness against the PACIO ToC Implementation Guide across real-world cases.
Duplicate & conflict reconciliation	Identified duplicate and conflicting entries and generated reconciled candidate information for review.	Conflict-detection precision and clinician agreement with AI-generated reconciliation suggestions.
Confidence & data-quality indicators	Generated confidence-scored candidate information using PIQI-informed data-quality logic.	Quantitative validation of confidence scoring and quality-assurance methods across broader clinical datasets.
Source traceability	Preserved source evidence and manifests so AI-assembled content could be traced back to originating artifacts.	Clinician use and perceived usefulness of source evidence during review, including its contribution to efficient adjudication.
Clinician review & human reconciliation	Provided a clinician-reviewable interface in which users could review AI-generated content, resolve conflicts, and determine readiness for downstream sharing.	Clinician agreement, usability, reconciliation consistency, and workflow fit in realistic care settings.
PACIO ToC FHIR output & conformance	Transformed harmonized data into clinician-reviewable FHIR bundles aligned with the PACIO ToC Implementation Guide.	Formal FHIR conformance validation across a broader set of cases and downstream systems.
Review burden & efficiency	Not quantitatively measured; the proof of concept demonstrated a workflow intended to reduce manual abstraction and chart review.	Time to review, manual abstraction effort, clinician burden, and performance relative to baseline workflows.
Real-world workflow integration	Defined an intended deployment architecture but did not evaluate production EHR or health information exchange integration or operational deployment.	Usability, workflow integration, operational feasibility, security, and controlled deployment in real post-acute care workflows.
Policy, regulatory & interoperability governance	Not evaluated as a proof-of-concept outcome; governance needs were identified as deployment considerations.	Assess applicable privacy, security, data-use, regulatory, federal interoperability, and exchange-governance requirements, including TEFCA where relevant.
Clinical & transition outcomes	Not measured; the proof of concept does not support claims of improved clinical outcomes or efficiency.	Transition quality and other clinically meaningful outcomes in real-world post-acute care settings.

Conclusions

This proof of concept demonstrates the feasibility of applying generative AI to structured and unstructured clinical data to produce reviewable, transition-relevant information. PACIO and FHIR provide the interoperability foundation, while AI enhances the usability of standards-based exchange through semantic structuring, relevance filtering, harmonization, and source traceability. Human oversight remains essential for clinical review, reconciliation, safety, and trust.

These conclusions are limited to technical feasibility and do not establish improvements in clinical outcomes, efficiency, or real-world workflow performance. Next steps include policy, regulatory, security, and interoperability-governance review; real-world validation; clinician usability studies; controlled pilot deployment; and benchmarking against baseline workflows. Future evaluation should determine whether the approach improves completeness, review efficiency, or transition quality. **The goal is not simply to exchange data, but to make interoperable data clinically meaningful, actionable, and ready for the next care team.**

Acknowledgements

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Contact & Call to Action

Collaborate with PACIO to evaluate and pilot AI-assisted, clinician-governed ToC workflows in real-world post-acute care settings.

Contact email: info@pacioproject.org

Website: www.PACIOProject.org

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